# lúbadh

## _travelers guide _for navigating time loops

| 2 | 10 | 4 |
|---|---|---|
| decks · stereo link | min per loop, per deck | taps per deck |

[ rec ] press  [ rec ] hold  [ shift ] = retrig/shift  opp. = the other deck's button  › then
◆ = preset / mode only

every control is per deck: left and right are identical

combos shown for one deck · mirror them for the other

amber = hold

### 01 transport — lbh_01

| | |
|---|---|
| [ rec ] | record › press again to stop & play. press while playing = **overdub** |
| [ erase ] | erase this deck's loop |
| [ shift ] | jump back to start |
| turn [ speed ] | speed & direction: center = stall, cw = forward to 4×, ccw = reverse to 4× |
| [ shift ] + turn [ speed ] | knob is muted until release, then jumps straight to the new speed |
| [ link ] up | right deck mirrors left · saves and loads stereo |
| [ shift ] | adds a playhead at start (up to 4) |
| ◆ multi-tap presets | |

### 02 recording — lbh_02

| | |
|---|---|
| [ erase ] + [ rec ] | **punch-in**: replaces audio while recording |
| [ shift ] [ shift ] + [ rec ] | **input monitoring**: on (rec lit) › armed (rec pulses) › off |
| armed [ rec ] | arm › record › stop & play · [ erase ] disarms, loop kept |
| [ shift ] + opp. [ rec ] | looping ← → **one-shot record** (records one pass, then stops) |
| [ rec ] jack | trig starts / stops recording, like [ rec ] (latching) |
| [ rec ] jack | gated: records only while the gate is high |
| ◆ rec jack = gated | |

* overdub keeps the old layer at 0.9 by default, so layers fade a little each pass (dub level, 04).

### __fig.01 : speed

white speed LEDs = notches




⅛ = octave down · 2× = octave up

### 03 playback — lbh_03

| | |
|---|---|
| [ shift ] + opp. [ erase ] | looping ← → **one-shot playback** (rec blinks white / mixed) |
| [ shift ] or [ retrig ] jack | one-shot: play one pass of start › length |
| touch the reel | warble or stall the tape · trimmer sets sensitivity |
| [ speed ] cv | ±5 v, summed with the knob |
| [ erase ] + opp. [ erase ] | flip playback direction · speed LEDs blink |
| ◆ v/oct speed mode only | |
| [ speed ] cv | tracks 1v/oct: sequence the loop's pitch |
| ◆ v/oct · seq presets | |

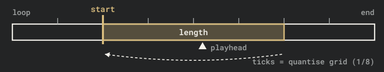
### 04 time knob — lbh_04

| | |
|---|---|
| [ shift ] + [ rec ] | cycle the time function (amber LED by [ clk out ] shows which) |
| LED off | **clock divisions**: triggers per loop at [ clk out ]: 1–12, 16, 24, 32, 64 |
| LED solid | **dub level**: how much old audio survives each overdub. high = delay feedback |
| LED pulsing | **user parameter**: crossfade, tape or slew (preset picks) |




### 05 start · length — lbh_05

| | |
|---|---|
| turn [ start ] | where the loop begins inside the recording |
| turn [ length ] | how much plays before jumping back. tiny = wavetable-like tones |
| [ shift ] + turn [ time ] | **quantise** start & length: lit LED sections = 1/2, 1/4, 1/8 … |
| [ shift ] + turn [ start ] | fine move that ignores the quantise grid (same for [ length ]) |




### 06 save · load — lbh_06

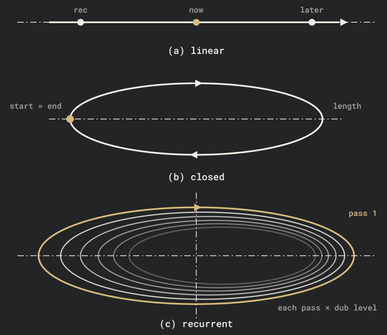
| | |
|---|---|
| [ shift ] [ erase ] + [ rec ] | **save** to usb › saved (folder 0, newest first) |
| [ shift ] + [ erase ] | **load menu** |
| in the menu | own [ time ] = loop · opposite [ time ] = folder · [ rec ] loads · hold [ shift ] previews 5 s · [ erase ] exits |
| [ link ] up | saves and loads stereo (mono loads of stereo are summed) |
| power-on | last session reloads automatically |

### 07 presets — lbh_07 · v2.1

| | |
|---|---|
| [ shift ] [ shift ] + [ erase ] | **preset menu** for that deck |
| turn [ time ] | pick 1 of 12 (shown on the LED ring) · [ rec ] loads · [ erase ] cancels |
| usb › presets | .txt files, sorted 01–12. build your own with the preset editor |

### __fig.02 : the return

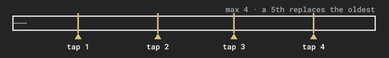
mm_02




### 08 multi-tap — lbh_08

◆ multi-tap presets only

| | |
|---|---|
| [ shift ] | new playhead from start |
| [ erase ] + [ shift ] | delete the taps |
| [ erase ] trig, cv exp. | removes taps remotely |




### 09 sync · clock — lbh_09

| | |
|---|---|
| [ clk out ] | triggers per loop (set by clock div) |
| [ retrig ] jack | resets the playhead to start: sync to an external clock |
| [ rec ] jack | external clock sets the loop length |
| ◆ clockable preset | |
| [ erase ] + [ rec ] | tap tempo instead of punch-in |
| ◆ erase + rec = tap tempo | |

### 10 aux · routing — lbh_10

| | |
|---|---|
| aux in fader | sends [ aux ] in to the left ← → right deck input (+ [ cv ] ±5 v) |
| aux out fader | blends left ← → right outputs at [ aux ] out (+ [ cv ] ±5 v) |
| rear switch | aux out fader pre or post the output level, per deck |




### 11 reset · calibrate — lbh_11

| | |
|---|---|
| [ shift ] [ shift ] + [ rec ] [ rec ] | **reset modes** to default · audio kept · speed LEDs pulse |
| power on, hold [ shift ]×2 [ erase ]×2 | **calibration**: speed knobs min, 0 v › [ rec ] each deck, then 1 v and 3 v; knobs max, no cv › [ rec ] = 5 v |
| amber blinks | voltage out of range: defaults used |

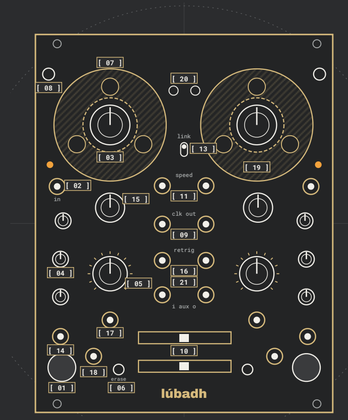
### __fig.03 : front panel

01–11 = sections · 13–21 = hardware

[ 01 ] __rec : record · overdub · stop
[ 02 ] __in : input + level
[ 03 ] __speed : rate & direction
[ 04 ] __time : clock div · dub · user
[ 05 ] __start : loop start
[ 06 ] __erase : erase · save / load
[ 07 ] __LED ring : position · presets
[ 08 ] __retrig/shift : restart · shift · taps
[ 09 ] __clk out : clock triggers
[ 10 ] __aux faders : in / out crossfade
[ 11 ] __speed cv : ±5 v · 1v/oct

[ 13 ] __link : stereo mirror
[ 14 ] __out : output + level
[ 15 ] __length : loop length
[ 16 ] __retrig : jack, restart
[ 17 ] __cv : aux fader cv
[ 18 ] __rec : jack, record gate
[ 19 ] __reel : capacitive touch
[ 20 ] __speed LEDs : notches
[ 21 ] __aux : in · out jacks

left deck labelled · right deck mirrors it




### _where things hide

12 · lbh_12

#### preset files

usb › presets › .txt · instruo preset editor

- speed record variable · fixed, play follow · hold
- speed control notched · stepped · smooth · v/oct, markers 0–4
- multi-tap engine off · on
- time user crossfade · speed slew · tape emulation
- rec jack latching · gated · external clock
- ext clock average, resolution, timeout, divisions
- quantise all · even · odd · powers of two
- erase + rec punch-in · tap tempo
- timing min loop, crossfade, retrig delay, max dub
- tape wow / flutter, crinkle, age, hysteresis, wear, saturation, reverb
- input filter low cut freq + q, high cut
- touch off · stall · dip

#### usb disk

- samples 12 folders × 12 loops
- saved folder 0, newest first
- presets first 12 show in the menu

#### tails & tags

250 ms, in the file name

- {s} silent tail (default): keeps timing
- {n} no tail: smoothest, timing loosens
- {c} copy tail: keeps timing, may click

#### hardware

- cv expander [ length ] [ start ] [ time ] cv + [ erase ] trig, per deck
- rear pcb touch on / off · pre / post output level
- front touch sensitivity trimmer

#### factory presets

- 01–05 tape looper · clean · classic · multi-tape · clockable
- 06–10 octave delay · seq mono · seq poly · melontron · broken tape